

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

8th Semester of B. Pharm. Examination

University Theory Examination April/May 2018

PH424 Dosage Form Design

Date: 10/05/2018, Thursday

Time: 10:00 a.m. to 1:00 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

20

1. What is the residence time of food in stomach in fasted condition?
 - A. 0.5-2 hr.
 - B. 2-3 hr.
 - C. 2.5-4.5 hr.
 - D. More than 4 hr.
2. _____ is used as plasticizer during the coating of tablet.
 - A. Polyethylene Glycol 4000
 - B. Polyethylene Glycol 1500
 - C. Polyethylene Glycol 6000
 - D. Polyethylene Glycol 400
3. From the following which dissolution apparatus can be used for the dissolution of multi particulate dosage form?
 - A. Basket
 - B. Paddle
 - C. Flow-Through Cell
 - D. Paddle over Disk
4. If dissolution media required to be changed during dissolution study, _____ dissolution apparatus is used.
 - A. Apparatus 1
 - B. Apparatus 3
 - C. Apparatus 5
 - D. Apparatus 7
5. Biorelevant dissolution cannot facilitate the selection of _____ properties of drug substance.
 - A. Salt forms
 - B. Polymorphs
 - C. Particle sizes
 - D. pH

6. Stability study can give the idea about the _____ of drug product.
 - A. Labelling instruction.
 - B. Dissolution
 - C. Hardness
 - D. Flow Properties
7. _____ Stability study is performed to predict the self-life of product.
 - A. Real time
 - B. Accelerated
 - C. Long-time
 - D. At room temperature
8. _____ as environmental factor affect the stability of drug product.
 - A. Moisture
 - B. pH
 - C. Leaching from container
 - D. Solubility
9. _____ is a most affecting variable for sedimentation of suspension.
 - A. Concentration of drug
 - B. Solubility of drug
 - C. Amount of preservative
 - D. Viscosity of medium
10. Prefomulation can't give the idea of _____.
 - A. Compatibility
 - B. Stability
 - C. Probable dosage form
 - D. Human dose
11. Amorphous form of drug can be identified using _____ method.
 - A. Particle size analysis
 - B. TEM
 - C. SEM
 - D. XRD
12. Excessive supersaturating modifies the _____ of drug.
 - A. Shape
 - B. Carr's index
 - C. Angle of repose
 - D. Precipitation.
13. From the following, which one is not affected by Particle size?
 - A. Density
 - B. Compressibility
 - C. pH
 - D. Dissolution
14. Magnesium stearate is not suitable with _____ of drug as lubricant.
 - A. Basic nature
 - B. Acidic nature
 - C. Neutral nature
 - D. Amphoteric nature
15. From the following which binder can be used in wet granulation and direct compression?
 - A. Potato starch
 - B. Polyvinyl Pyrrolidone
 - C. Pregelatinized Starch
 - D. Polyethylene Glycol

16. _____ is used as inorganic diluent in tablet formulation.
- Lactose
 - Calcium phosphate
 - Sorbitol
 - Mannitol
17. What is the mechanism of disintegration of tablet contain sodium starch glycolate?
- Gas release
 - Enzymatic
 - Swelling
 - Capillary action
18. Kollidon is used as _____ in tablet formulation.
- Binder
 - Disintegrations
 - Lubrication
 - pH modifier
19. What is the used of Colloidal silicon dioxide in the suspension?
- Surfactant
 - Solubility enhancement
 - Viscosity modifier
 - Particles size reduction.
20. _____ is used as co-solvent in parenteral formulation.
- Propylene glycol
 - Water
 - Tween 80
 - Span 20

SECTION – II

Q 2 Attempt any TWO of the following

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|----------|--|-----------|
| A | Define the polymorphism. What is the significance of polymorphism in formulation developments? | 05 |
| B | Explain the key steps of design of experiments. | 05 |
| C | What is overage? What is permissible limit? Describe calculation of overages with example. | 05 |

Q 3 Attempt any TWO of the following

- | | | |
|----------|---|-----------|
| A | Define preformulation. Explain the steps for preformulation study for development of product. | 05 |
| B | Enlist the types of design of experiments. Give the application and justification of used of each type. | 05 |
| C | Describe the techniques used to ascertain Drug-Excipient Compatibility study. | 05 |

SECTION – III

Q 4 Attempt any FOUR of the following

- A** A drug in suspension (100mg/ mL) degrades by zero-order kinetics with a rate constant of 0.5 mg/mL hr. Calculate the shelf life of drug. How much drug will be left after 48 hr? **05**
- B** What are physicochemical factors affecting preformulation study? Discuss the importance of crystal structure in preformulation. **05**
- C** Discuss the requirement related to stability testing with emphasizing on matrixing/ bracketing technique. **05**
- D** How is accelerated stability study carried out? How the results of it can be correlated with real time study? **05**
- E** Discuss the significance of particle size and shape in formulation, stability and bioavailability. **05**
- F** What are the stability guidelines as per ICH? Write a note on Photo stability study. **05**

Q 5 Attempt any FOUR of the following

- A** How dissolution data are compared? Describe Dissolution profile comparison based on f_1 and f_2 and give their equation. **05**
- B** Enumerate additives used in tablets. Discuss antifrictional agents, binders and disintegrating agents in detail. **05**
- C** What is intrinsic dissolution rate? Write a note on factors affecting dissolution. **05**
- D** Enumerate the compendial apparatus for dissolution studies. Describe the dissolution apparatus-I of Indian Pharmacopoeia. **05**
- E** Describe the ideal properties of a preservative. Describe in detail about antimicrobial preservatives. **05**
- F** Explain in detail about the diffusion layer model theory of dissolution. **05**